WORKED SOLUTION

STEP I 1991, Question 5

Pure • Regular Sets of Vectors

> **STEP I 1991, Question 5**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/91-s1-q5

The key idea

Condition (ii) says the vectors add up to $\mathbf{0}$, and the cheapest way to turn a vector equation into a number is to take a dot product with one of the vectors, which condition (i) tells us how to evaluate. After that, the whole question is about counting: a vector whose dot products with some fixed vectors are all prescribed has very few possible positions, so there's only room for a small number of vectors in the set.

Solution

The identity $(n-1)\beta = -\alpha^2$

By (ii), $\mathbf{a}_1 + \mathbf{a}_2 + \cdots + \mathbf{a}_n = \mathbf{0}$. Take the dot product of both sides with $\mathbf{a}_1$. By (i), the term $\mathbf{a}_1 \cdot \mathbf{a}_1$ is α^2 and each of the other $n-1$ terms $\mathbf{a}_1 \cdot \mathbf{a}_j$ $(j \neq 1)$ is β, so

$$\alpha^2 + (n-1)\beta = 0, \qquad \text{that is,} \qquad (n-1)\beta = -\alpha^2,$$

as required. In particular $\beta = -\alpha^2/(n-1) < 0$.

Two dimensions

Now $\alpha^2 = \mathbf{a}_1 \cdot \mathbf{a}_1 = 1$, so every $\mathbf{a}_i$ is a unit vector, and

$$\beta = -\frac{1}{n-1}.$$

Write $\mathbf{a}_i = \begin{pmatrix} x_i \\ y_i \end{pmatrix}$ for $i \geq 2$. The condition $\mathbf{a}_1 \cdot \mathbf{a}_i = \beta$ says $x_i = \beta$, and then $|\mathbf{a}_i| = 1$ forces $y_i = \pm\sqrt{1-\beta^2}$. So every vector other than $\mathbf{a}_1$ is one of

$$\begin{pmatrix} \beta \\ \sqrt{1-\beta^2} \end{pmatrix}, \qquad \begin{pmatrix} \beta \\ -\sqrt{1-\beta^2} \end{pmatrix}.$$

Ask yourself How does knowing the candidates limit n?

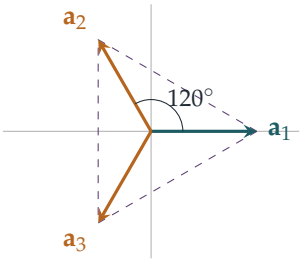
The vectors in the set are distinct, so $\mathbf{a}_2, \ldots, \mathbf{a}_n$ are $n-1$ *different* candidates. There are at most two candidates, so $n-1 \leq 2$, i.e. $n \leq 3$. Since $n \geq 2$, either $n = 2$ or $n = 3$.

$n = 2$. Then $\beta = -1$, so the two candidates coincide and $\mathbf{a}_2 = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$. This works: $\mathbf{a}_1 \cdot \mathbf{a}_2 = -1 = \beta$ and $\mathbf{a}_1 + \mathbf{a}_2 = \mathbf{0}$.

$n = 3$. Then $\beta = -\frac{1}{2}$ and $\sqrt{1-\beta^2} = \frac{\sqrt{3}}{2}$, and $\mathbf{a}_2, \mathbf{a}_3$ must be the two different candidates:

$$\begin{pmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -\frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}$$

(in either order). Check: $\mathbf{a}_2 \cdot \mathbf{a}_3 = \frac{1}{4} - \frac{3}{4} = -\frac{1}{2} = \beta$, and the three vectors sum to $\mathbf{0}$. They point to the vertices of an equilateral triangle centred at the origin.



(Another way to pin down β: for $n \geq 3$ we have $-1 < \beta < 0$, so the two candidates really are different and $\mathbf{a}_2, \mathbf{a}_3$ must be both of them. Then $\mathbf{a}_2 \cdot \mathbf{a}_3 = \beta^2 - (1-\beta^2) = 2\beta^2 - 1$, and setting this equal to β gives $(2\beta+1)(\beta-1) = 0$. So $\beta = -\frac{1}{2}$, which again means $n = 3$.)

Three dimensions

Again $\alpha = 1$ and $\beta = -1/(n-1)$. The case $n = 2$ is as before: $\beta = -1$, and the only unit vector with $\mathbf{a}_1 \cdot \mathbf{a}_2 = -1$ is $\mathbf{a}_2 = -\mathbf{a}_1 = (-1, 0, 0)^{\mathrm{T}}$, which lies in the x-y plane. So $\{\mathbf{a}_1, -\mathbf{a}_1\}$ is one answer.

Now suppose $n \geq 3$, so $-\frac{1}{2} \leq \beta < 0$. Since $\mathbf{a}_2$ lies in the x-y plane, the same argument as in two dimensions gives

$$\mathbf{a}_2 = \begin{pmatrix} \beta \\ \varepsilon\sqrt{1-\beta^2} \\ 0 \end{pmatrix}, \qquad \varepsilon = \pm 1,$$

and the y-component $y_2 = \varepsilon\sqrt{1-\beta^2}$ is non-zero because $\beta \neq \pm 1$.

Ask yourself What do the conditions say about $\mathbf{a}_i$ for $i \geq 3$?

Write $\mathbf{a}_i = (x_i, y_i, z_i)^{\mathrm{T}}$ for $i \geq 3$. Then

$$\begin{aligned} \mathbf{a}_1 \cdot \mathbf{a}_i = \beta &\implies x_i = \beta, \\ \mathbf{a}_2 \cdot \mathbf{a}_i = \beta &\implies \beta^2 + y_2 y_i = \beta \implies y_i = \frac{\beta(1-\beta)}{y_2}, \\ |\mathbf{a}_i| = 1 &\implies z_i^2 = 1 - \beta^2 - y_i^2. \end{aligned}$$

So x_i and y_i are the same for every $i \geq 3$, and z_i can only be one of the two square roots. As the vectors are distinct, there are at most two of $\mathbf{a}_3, \ldots, \mathbf{a}_n$, so $n - 2 \leq 2$ and $n \leq 4$.

$n = 3$. By (ii), $\mathbf{a}_3 = -\mathbf{a}_1 - \mathbf{a}_2$, which lies in the x-y plane. So the set is a regular set in the plane $z = 0$, and by the two-dimensional result it is

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \\ 0 \end{pmatrix}, \quad \begin{pmatrix} -\frac{1}{2} \\ -\frac{\sqrt{3}}{2} \\ 0 \end{pmatrix},$$

with $\mathbf{a}_2$ either of the last two. (The formulas above agree: $\beta = -\frac{1}{2}$ gives $y_3 = -y_2$ and $z_3^2 = 1 - \frac{1}{4} - \frac{3}{4} = 0$.)

$n = 4$. Now $\beta = -\frac{1}{3}$, so $\sqrt{1-\beta^2} = \frac{2\sqrt{2}}{3}$ and $y_2 = \varepsilon\frac{2\sqrt{2}}{3}$. Then

$$y_i = \frac{(-\frac{1}{3})(\frac{4}{3})}{\varepsilon\frac{2\sqrt{2}}{3}} = -\varepsilon\frac{2}{3\sqrt{2}} = -\varepsilon\frac{\sqrt{2}}{3}, \qquad z_i^2 = 1 - \frac{1}{9} - \frac{2}{9} = \frac{2}{3}.$$

Since $\mathbf{a}_3 \neq \mathbf{a}_4$, they must take the two opposite values $z = \pm\sqrt{2/3} = \pm\frac{\sqrt{6}}{3}$. This gives

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} -\frac{1}{3} \\ \varepsilon\frac{2\sqrt{2}}{3} \\ 0 \end{pmatrix}, \quad \mathbf{a}_3, \mathbf{a}_4 = \begin{pmatrix} -\frac{1}{3} \\ -\varepsilon\frac{\sqrt{2}}{3} \\ \pm\frac{\sqrt{6}}{3} \end{pmatrix},$$

one set for each choice $\varepsilon = \pm 1$. We have only used the conditions involving $\mathbf{a}_1$ and $\mathbf{a}_2$, so we must still check the rest. First, $\mathbf{a}_3 \cdot \mathbf{a}_4 = \frac{1}{9} + \frac{2}{9} - \frac{6}{9} = -\frac{1}{3} = \beta$. Second, the sum: the x-components give $1 - 3 \cdot \frac{1}{3} = 0$, the y-components give $\varepsilon\left(\frac{2\sqrt{2}}{3} - \frac{\sqrt{2}}{3} - \frac{\sqrt{2}}{3}\right) = 0$, and the z-components cancel. So both sets are regular. Geometrically, the four vectors point to the vertices of a regular tetrahedron centred at the origin.

Summary. The regular sets are: $\{\mathbf{a}_1, -\mathbf{a}_1\}$ ($n = 2$); the equilateral triangle in the x-y plane ($n = 3$); and the two regular tetrahedra above ($n = 4$, $\varepsilon = \pm 1$, mirror images in the x-z plane). No regular set has $n \geq 5$.

Checks

In each set the vectors should sum to zero and all pairwise dot products should be equal. For the tetrahedron, the angle between any two vectors is $\cos^{-1}(-\frac{1}{3}) \approx 109.47°$, the familiar bond angle of methane, and the identity gives $(4-1)(-\frac{1}{3}) = -1 = -\alpha^2$. We checked all five sets numerically (Gram matrix and centroid to 10^{-12}), and a brute-force search in the plane found regular sets only for $n = 2$ and $n = 3$, for n up to 11.

Common slips

- Forgetting that the vectors must be *distinct*: this is the condition that turns "at most two candidates" into a bound on n.
- In the $n = 4$ case, finding $\mathbf{a}_3, \mathbf{a}_4$ from their dot products with $\mathbf{a}_1, \mathbf{a}_2$ and then not checking $\mathbf{a}_3 \cdot \mathbf{a}_4 = \beta$ and the centroid condition.
- Missing the $n = 2$ case or the choice of sign ε in three dimensions.

Beyond STEP

These sets are the vertices of the regular simplex: n points in $(n-1)$-dimensional space, all the same distance apart. The matrix of dot products $G = (\mathbf{a}_i \cdot \mathbf{a}_j)$ (the *Gram matrix*) is $(\alpha^2 - \beta)I + \beta J$, where J is the all-ones matrix. Its rank is $n - 1$, and the rank of a Gram matrix can't exceed the dimension of the space, which gives $n \leq d + 1$ in d dimensions at once. Strang's *Introduction to Linear Algebra* covers rank, dot products and Gram matrices.

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